Discuss the kinetics of free radical polymerization.

~~Also derive the expression for the~~ average kinetic chain length for anionic polymerization.

Kabir ele-46

Kinetics for free radical polymerization:

1. Initiation.

$$I \xrightarrow{K_d} 2R\cdot \quad \text{Radical generation}$$

$$R\cdot + M \xrightarrow{K_i} M_1\cdot \quad \text{Initiation.}$$

Assuming $K_i >> K_d$. Therefore the rate of initiation, R_i is given by -

$$R_i = \frac{d[M_1\cdot]}{dt} = 2fK_d[I] \quad \text{---(i)}$$

where f = efficiency of the ~~inhibitor~~ initiator.

2. Propagation:

Assuming that the reactivity of the g

$M_1^{\cdot} + M \xrightarrow{k_p} M_2^{\cdot}$

$M_2^{\cdot} + M \xrightarrow{k_p} M_3^{\cdot}$

$M_3^{\cdot} + M \xrightarrow{k_p} M_4^{\cdot}$

$M_x^{\cdot} + M \xrightarrow{k_p} M_{x+1}^{\cdot}$

Assuming that, the reactivity of the growing chain is independent of the length of the chain.

Therefore,

The rate of propagation —

$$R_p = -\frac{d[M]}{dt} = k_p [M^{\cdot}][M] \quad \text{---(ii)}$$

3. Termination:

$M_x^{\cdot} + {}^{\cdot}M_y \xrightarrow{k_{tc}} M_x - M_y$ (combination)

$M_x^{\cdot} + {}^{\cdot}M_y \xrightarrow{k_{td}} M_x + M_y$ (Disproportionation)

since two radicals are consumed in every termination—

$$R_t = 2k_t [M^{\cdot}]^2 \quad \text{---(iii)}$$

Steady state approximation.

In polymerization, the concentration of radicals becomes constant because, $R_i = R_t$.

$$\therefore 2fk_d[I] = 2k_t[M\cdot]^2$$

$$\Rightarrow [M\cdot] = \left(\frac{fk_d[I]}{k_t}\right)^{1/2} \quad \text{(ii)}$$

substituting the value of $[M\cdot]$ in eqn (ii)

$$R_p = k_p[M\cdot][M]$$

$$= k_p[M]\left(\frac{fk_d[I]}{k_t}\right)^{1/2}$$

This is the rate expression for free radical polymerization.//

Average kinetic chain length:

The average number of monomer units polymerized per chain initiated.

Thus, $\bar{\upsilon} = \frac{R_p}{R_t}$

$$= \frac{k_p[M\cdot][M]}{2k_t[M\cdot]^2} = \frac{k_p[M]}{2k_t[M\cdot]}$$

17 Deduce Carothers equation.*

Carothers equation:

Consider a flask of monomer. If no of initially reacted there are N_0 molecules in the flask at time zero and N at time t is no of molecules remaining at time t after reaction. Then, degree of polymerization $\bar{x}_n$ the average at time t is given by $\frac{N_0}{N}$.

Thus, $\bar{x}_n = \frac{N_0}{N}$ ....(i)

since, the amount reacted is $N_0 - N$

Thus the extent of reaction — conversion

$$p = \frac{\text{No. of group which have reacted}}{\text{no. of group initially present.}}$$

$$\Rightarrow p = \frac{N_0 - N}{N_0}$$

$$\Rightarrow p = 1 - \frac{N}{N_0}$$

$$\Rightarrow \frac{N}{N_0} = 1 - p$$

$$\Rightarrow \frac{N_0}{N} = \frac{1}{1-p} \quad \text{....(ii)}$$

...m eqn (i) and (ii)

the average degree of polymerization

$$\bar{x}_n = \frac{1}{1-p}$$

This is known as Carothers equation.

Discuss the kinetics of step-reaction polymerization in presence and absence of acid catalyst.

Kinetics of step reaction polymerization:
in absence of acid catalyst:

Dibasic acid + alcohol (glycol) $\xrightarrow[\text{of acid catalyst}]{\text{in absence}}$ polyester

The rate of reaction —

$$-\frac{d[COOH]}{dt} = -\frac{d[OH]}{dt} = K[COOH]^2[OH] \quad \text{---- (i)}$$

If the concentration of carboxyl and hydroxyl groups are same i.e.

$[OH] = [COOH] = c$, then eqn (i) becomes -

$$-\frac{dc}{dt} = Kc^3 \quad \text{---- (ii)}$$

$$\Rightarrow -\frac{dc}{c^3} = dt * K \quad \text{---- (iii)}$$

Integrating eqn (iii) ~~we get~~ using limit $0 \to t$ and $C_0 \to C$

$$-\int_{C_0}^{C} \frac{1}{C^2}\, dC = K\int_0^t dt$$

$$\Rightarrow -\left[\frac{C^{-2}}{-2}\right]_{C_0}^{C} = Kt$$

$$\Rightarrow \frac{1}{2}\left(\frac{1}{C^2} - \frac{1}{C_0^2}\right) = Kt$$

$$\Rightarrow 2Kt = \frac{1}{C^2} - \frac{1}{C_0^2} \quad \ldots\ldots\ldots (iv)$$

let p is defined as fraction of functional group which undergoes esterification at time t. and is given as —

$$p = \frac{C_0 - C}{C_0}$$

$$\Rightarrow p = 1 - \frac{C}{C_0}$$

$$\Rightarrow \frac{C}{C_0} = 1 - p$$

$$\Rightarrow C = C_0(1-p) \quad \ldots\ldots\ldots (*)$$

ing the value of c in eqn (iv) –

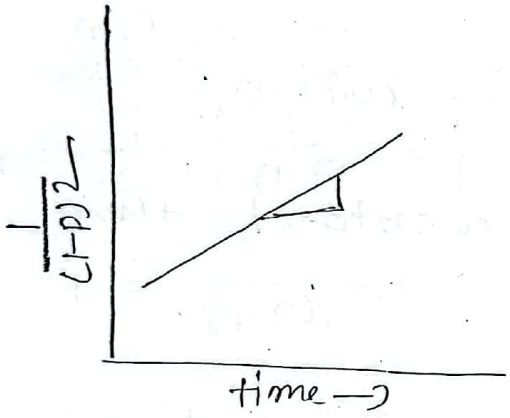
$$2kt = \frac{1}{c_0^2(1-p)^2} - \frac{1}{c_0^2}$$

$$\Rightarrow 2kt = \frac{1}{c_0^2}\left\{\frac{1}{(1-p)^2} - 1\right\}$$

$$\Rightarrow 2c_0^2 kt = \frac{1}{(1-p)^2} - 1 \quad \text{---------- (v)}$$

$$\Rightarrow \frac{1}{(1-p)^2} = 1 + 2c_0^2 kt \quad \text{---------}$$

Hence the plot of $\frac{1}{(1-p)^2}$ vs time, t gives a straight line



Again,

Degree of polymerization

$$DP = \frac{c_0}{c} = \frac{1}{(1-p)}$$

$$\Rightarrow DP^2 = \frac{1}{(1-p)^2} = 2kc_0^2 t + 1 \quad \text{---- (*)}$$

From (*) Dp appears as the second power and time is first power and hence the polymer molecular weight increase only slowly.

2017 4(b) ✗

In presence of acid catalyst :-

Acid + alcohol $\xrightarrow[\text{of acid catalyst}]{\text{in presence}}$ polyester.

The rate of this reaction —

$$-\frac{d[COOH]}{dt} = k\,[\text{catalyst}][COOH][OH] = -\frac{d}{dt}[OH] \quad \text{..... (vi)}$$

when

$$[OH] = [COOH] = c$$

Eqn (vi) becomes -

$$-\frac{dc}{dt} = k\,[\text{catalyst}]\,c^2 \quad \text{..... (vii)}$$

since catalyst is a constant, then equation (vii) becomes —

$$-\frac{dc}{dt} = k_x' c^2$$

$$\Rightarrow -\frac{dc}{c^2} = k_x\,dt \quad \text{..... (viii)}$$

Integrating eqn (viii) using limit $t \to 0$ and $c_0 \to c$

$$\int_{C_0}^{C} \frac{1}{C^2}\, dC = K \int_0^t dt$$

$$\Rightarrow -\left[\frac{C^{-1}}{-1}\right]_{C_0}^{C} = K_{2c} t$$

$$\Rightarrow \left(\frac{1}{C_0} - \frac{1}{C_0}\right) = K_{2c} t$$

$$\Rightarrow K_{2c} t = \frac{1}{C} - \frac{1}{C_0} \quad \text{---------- (ixc)}$$

Putting $C = C_0(1-P)$ in eqn (ixc)

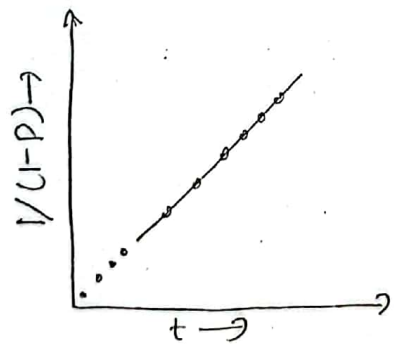
$$K_{2c} t = \frac{1}{C_0(1-P)} - \frac{1}{C_0}$$

$$\Rightarrow K_{2c} t = \frac{1}{C_0}\left\{\frac{1}{(1-P)} - 1\right\}$$

$$\Rightarrow K_{2c} C_0 t = \frac{1}{(1-P)} - 1$$

$$\therefore \left(\frac{1}{1-P}\right) = K_{2c} C_0 t + 1$$

Thus a plot of $\frac{1}{1-P}$ vs t gives a straight line.



Degree of polymerization.

$$Dp = \frac{C_0}{C} = \frac{1}{1-p}$$

$$\Rightarrow Dp = K_{2e} C_0 t + 1 \quad \text{---} (**)$$

From (**) Dp appears proportional with time. This means that the MW built up in this case will be faster. Hence 2nd pathway is more efficient.

What is meant by inhibition and retardation? Mention one use of inhibitions

Retardation:

Retardars are chemical substance that can react with a radical to form stable products which have no capability of adding monomer is called retardation. This process is known as retardation.

Inhibition:

If the retardar is every effective, capable of inhibiting ~~of killing of~~ polymer chain growth and no polymer can form in this condition, then it is called inhibition.

Use of inhibitors:

i) Reduce the conc of radicals

ii) Shortend the average ~~life time~~ of radical and thus length of polymer chain.

2017
4(c)
2014

From the statistical analysis of distribution of molecular weight of step reaction polymerization, derive expression for the number distribution function and weight distribution function?

Let p is the probability that a functional group reacted at a time t. Thus the finding of functional group arranged is (1-p).

Now, let a given molecule contains x repeating units and such molecule contains (x-1) reacted functional group.

Thus,

probability of (x-1) functional group is $= p^{x-1}$

probability of unreacted group $= (1-p)$

therefore. total probability for complete molecule $= P^{x-1}(1-P)$.

If there are N no. of molecules, total no of x-mers is given by.

$$N_x = NP^{x-1}(1-P) \quad \text{---- (i)}$$

If the total no. of unit present is No, then

$$N = N_o(1-P)$$

Now eqn (i) becomes Σ

$$\Rightarrow N_x = N_o(1-P)P^{x-1} \quad \text{---- (ii)}$$

This is the no. distribution function for a linear stepwise polymerization.

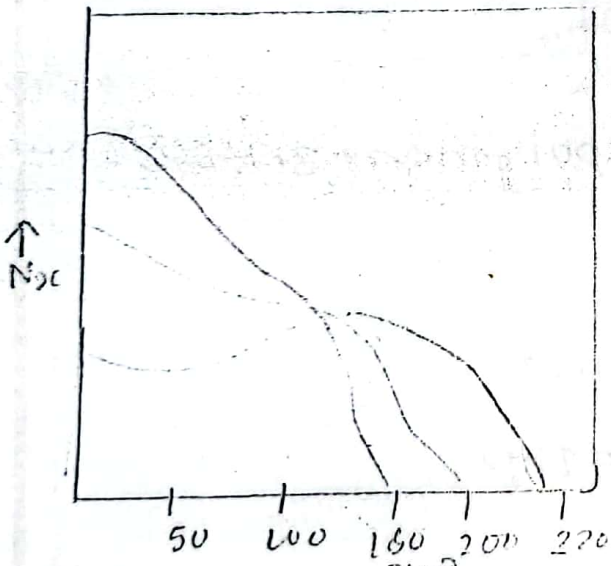


Fig: Number fraction distribution for linear step polymerization

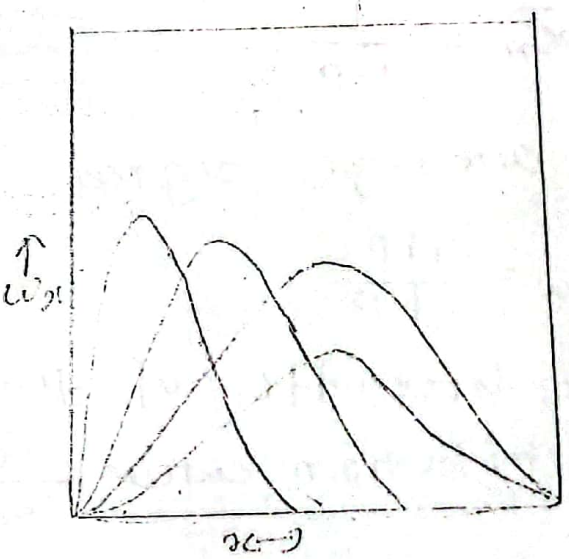


Fig: wt. fraction distribution in linear step polymerization

Again the weight distribution function w_x of x-mers is given by—

$$w_x = \frac{x N_x}{N_0}$$

$$= \frac{x N_0 (1-p)^2 p^{x-1}}{N_0} \quad [\text{using eqn (ii)}]$$

$$= x(1-p)^2 p^{x-1}$$

This is wt. distribution function for a linear stepwise polymerization.

———o———

No average degree of polymerization -

$$\bar{x}_n = \frac{1}{1-p}$$

wt. average degree of polymerization

$$\bar{x}_w = \frac{1+p}{1-p}.$$

Now, breadth of the M.W distribution curve $\frac{\bar{x}_w}{\bar{x}_n} = 1+p$.

Q16
4(c)

would you limit molecular weight in step-growth polymerisation.

Limitting of molecular weight in step-growth polymerizations—

1. To stop the polymerization reaction by rapid cooling when the desired M.w is attained.
2. To use an excess amount of one monomer.
3. To use small amount of monofunctional reactant.

Kakie

(d) Mention the conditions necessary for higher molecular weight in step growth polymerization.

Molecular weight in step growth polymerization is determined by the fractional conversion p of the monomer during polymerization.

A The conditions necessary for higher molecular weight in step growth polymerization —

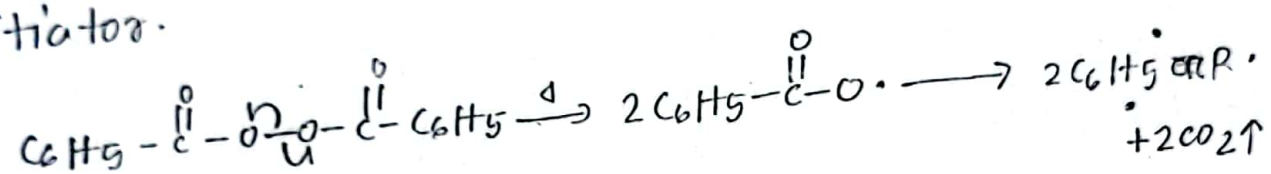
1. High conversation
2. High monomer purities
3. High reaction yields
4. Stoichiometric equivalance of functional group.

Explain the Initiation, propagation and termination steps with suitable examples for free radical polymerization.

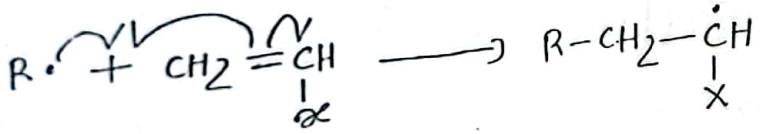
Free radical polymerization:

Initiation:

Initiation of a chain reaction involves the formation of free radicals from an initiator. Benzoyl peroxide is commonly used initiator.

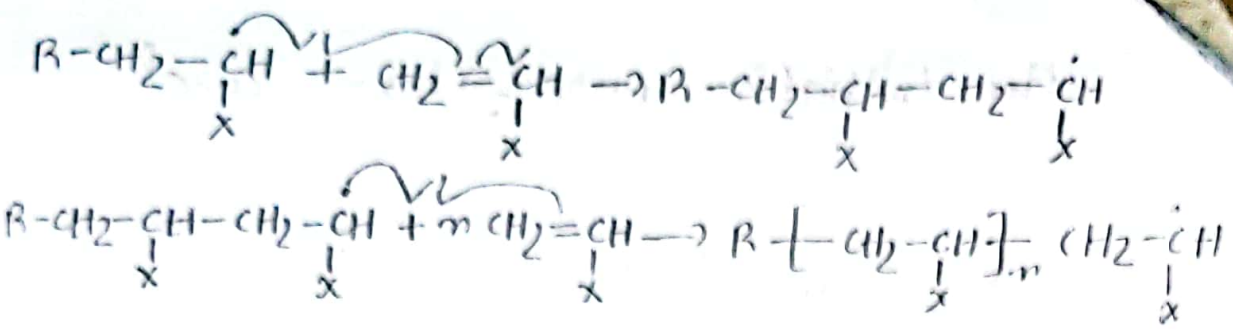


or, Peroxide → Radicals (R·)



Propagation:

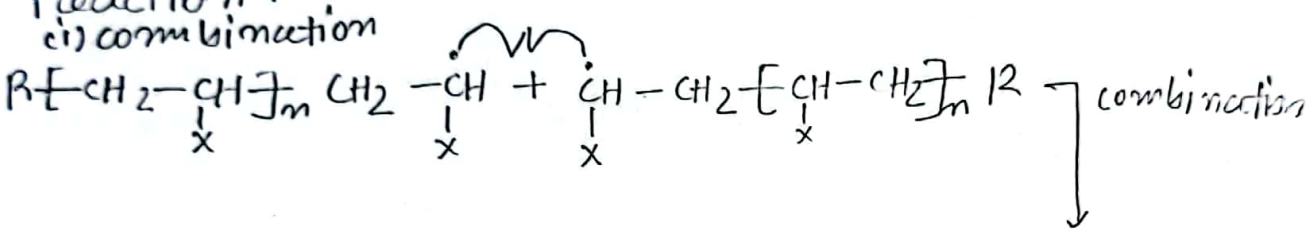
In propagation unit, the reaction site of the first monomer unit attacks the double bond of a fresh monomer.

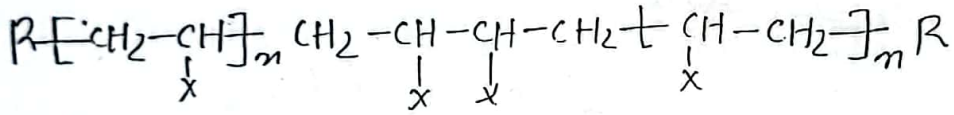


Termination:

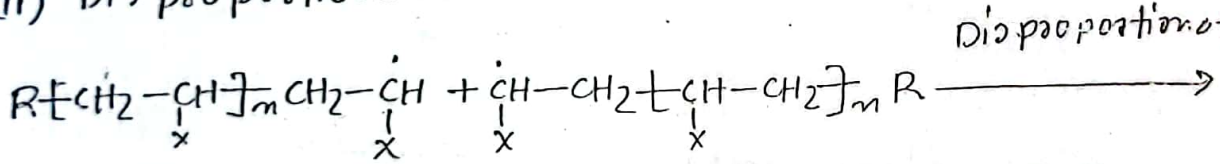
Termination can take place either by the combination of two free radical chain to form one molecule or by transfer of a hydrogen atom in a disproportionation reaction.

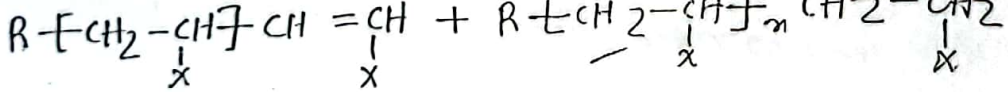
(i) combination





(ii) Disproportionation





reaction polymerization	Radical chain polymerization
1) A step reaction polymerization is a stepwise reaction in which a high-molecular weight polymer is formed after a large no. of steps.	1) A radical chain polymerization is a chain growth polymerization in which the kinetic chain carriers are radicals.
2. Monomers consumed rapidly while M.W. increases slowly.	2. Monomers consumed relatively slowly but M.W increased rapidly.
3. It is condensation polymerization.	3. It is addition polymerization
4. No initiator needed.	4. Initiator is essential for this polymerization
5. No termination step, end group still reactive. 6. DP low to moderate	5. Usually chain terminating step involved. 6. DP may be high.

Define the term autoacceleration. 2015. 5(c)

The increase in polymerisation and molecular weight of certain vinyl monomers during bulk polymerisation is known as autoacceleration

Auto acceleration only occur during polymerisation with high concentration of monomer. The reaction progress normally for a while. Then suddenly the rate of polymerisation goes up dramatically due to auto acceleration, The molecular chain that grows during the accelerated period is substantially higher than that of chains that grew earlier

What is ceiling temperature?

Ceiling temperature (T_c) is a measure of the tendency of polymers to revert their monomers. when a polymer is at it's ceiling temperature, the rate of polymerisation and depolymerisation are equal.

The temperature rating of polymerisation and depolymerisation –

$$k_i = A_i e^{-E_i/RT}$$

$$k_p = A_p \cdot e^{-E_p/RT}$$

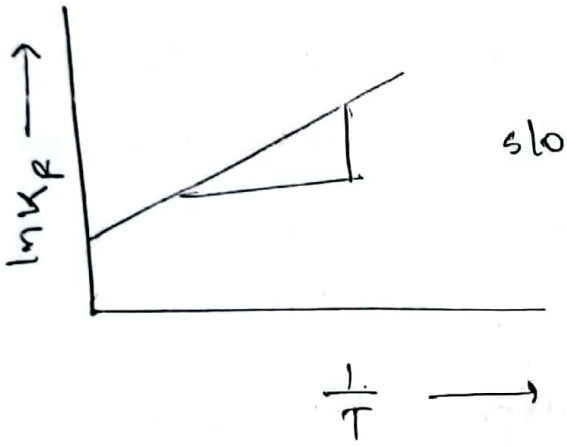
$$k_t = A_t \cdot e^{-E_t/RT}$$

overall rate constant

$$k_R = \frac{k_i \cdot k_p}{k_t} = \frac{A_i \cdot A_p}{A_t} \cdot \frac{e^{-E_i/RT} \cdot e^{-E_p/RT}}{e^{-E_t/RT}}$$

$$\Rightarrow k_R = \frac{A_i \cdot A_p}{A_t} \cdot e^{-(E_i + E_p - E_t)/RT}$$

$$\Rightarrow \ln k_R = \frac{E_t - E_i - E_p}{RT} + \ln \frac{A_i A_p}{A_t}$$



$$\text{slope} = \frac{E_t - E_i - E_p}{R}$$

(*)

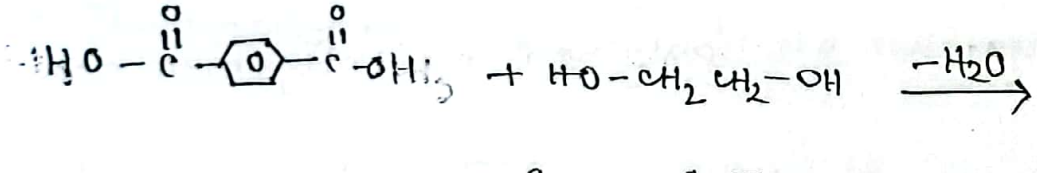
- polymerisation
  - Step growth or condensation polymerisation
  - Addition polymerisation
    - free radical polymerisation
    - ionic polymerisation
      - cationic polymerisation
      - anionic polymerisation

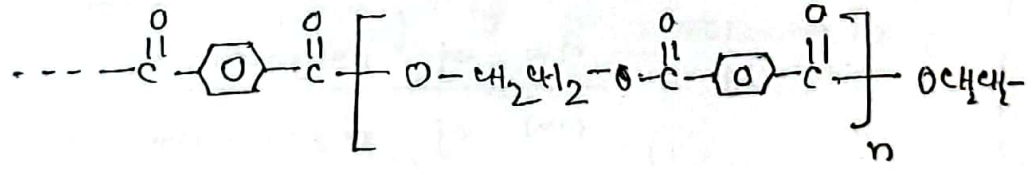
⊛ Step growth or condensation polymerisation:-

A condensation takes place between two bifunctional molecules to produce one larger polyfunctional molecules with the possible elimination of a small molecule such as H_2O, NH_3, HCl etc. the reaction continues until one of the reagents is used up.

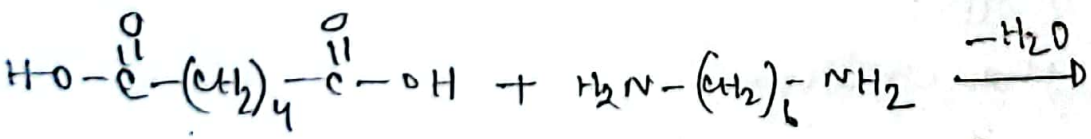
common types —
1) polyesters
2) polyamides
3) polycarbonate
4) polyurethanes

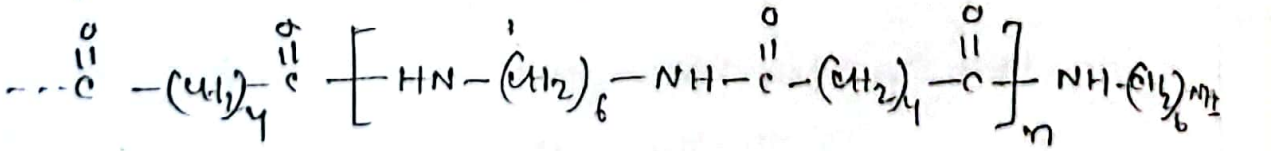
Polyesters: polymer of terephthalic acid and ethylene glycol, made by the trans-esterification of methyl ester





Polyamides: usually made from reaction of diacids with diamines.





* MW in step growth polymerisation :-

Polymer MW in step growth polymeration is determined by the fractional conversion p, of the monomer during the polymerisation.

Extent of reaction p :-

$$p = \frac{\text{number of functional groups that have reacted}}{\text{number of functional group originally present}}$$

$$= \frac{N_0 - N}{N_0}$$

* Degree of polymerisation (DP) :-

The degree of polymerisation or DP is usually defined as the number of monomeric units in a macromolecule or polymer or oligomer molecul

$$DP = \frac{\text{MW of polymer}}{\text{MW of monomer}}$$

→ One way to express M.W. is the degree of polymerination - number of repeating units in the chain.

ionic polymerisation:

Ionic polymerination is a particular type of addition polymerisation involving with carbonium ion and carbanion as the chain carries forming high molecular weight at an extremely high rate. ionic polymerasition also involves initiation, propogation and termination.

classification:-

1) cationic polymerisation
2) Anionic polymerisation

* Thermodynamics of free radical polymerisation:

The Gibbs free energy of polymerisation,

$$\Delta G_p = \Delta H_p - \Delta T \Delta S_p \quad \text{——(1)}$$

where ΔH_p is called the heat of polymerisation defined as

$$\Delta H_p = E_{dp} - E_p \quad \text{——(2)}$$

where E_{dp} and E_p are activation energy of depolymerisation and polymerisation.

At low or moderate temperature ΔG_p is negative.

As temperature increase ΔG_p become less negative.

At celling temperature (T_c) the polymerisation reaches equilibrium. In other words rate of polymerisation and depolymerisation become equal and

$$\Delta G_p = 0$$

The celling temperature then obtained from eqn (1) as —

$$T_c = \Delta H_p / \Delta S_p$$

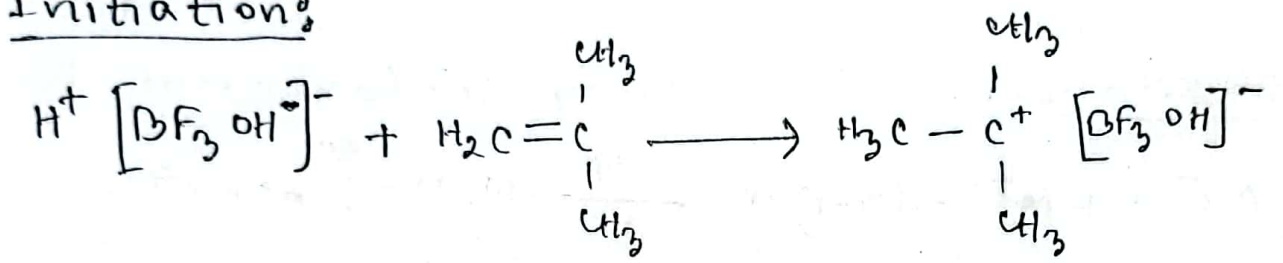
Mechanism of cationic polymerisation :-

catalyst : lewis acid

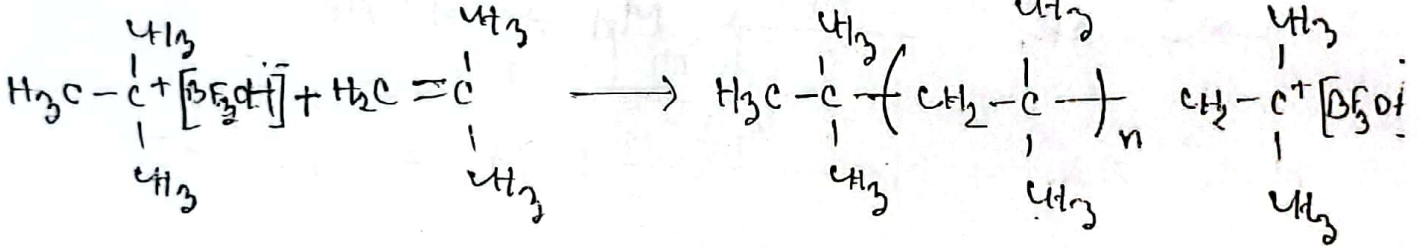
Co-catalyst : water

$$BF_3 + H_2O \rightleftharpoons H^+ [BF_3OH^-]^-$$

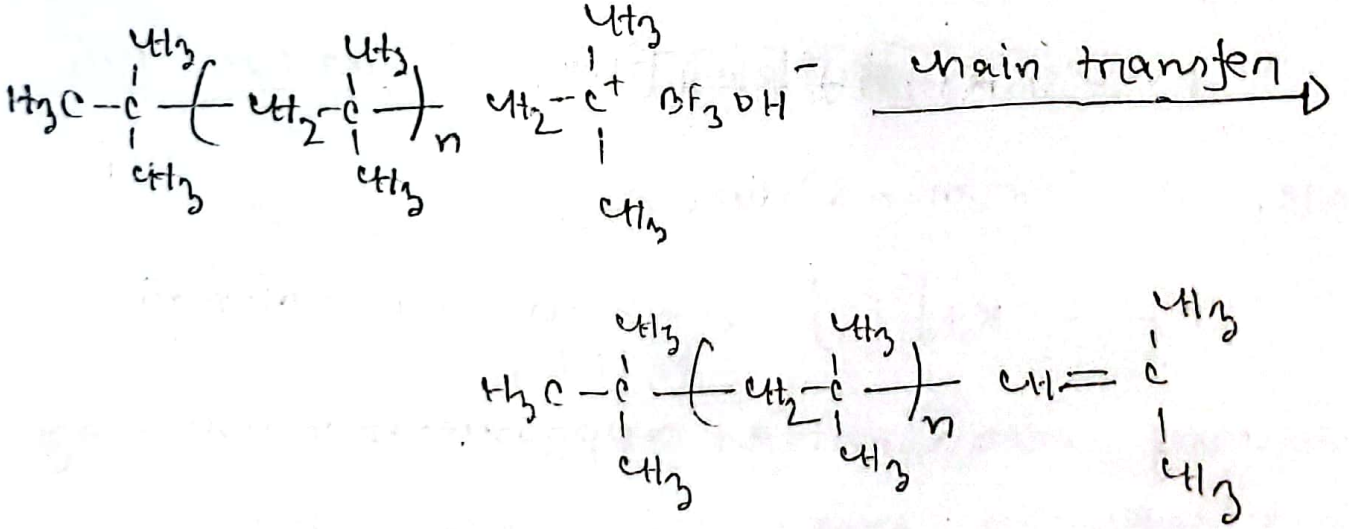
Initiation:



Propogation:



Termination :



2019-20

Kinetics of cationic polymerization:-

Let A be the catalyst and RH co-catalyst.

Initiation:

$$A + RH \xrightarrow{K} H^+ + AR^-$$

$$H^+ + AR^- + M \xrightarrow{K_i} HM^+ + AR^-$$

Propogation:

$$AR^- + HM^+ + (n-1)M \xrightarrow{K_p} HM_n^{\oplus} + AR^{\ominus}$$

Termination:

$$HM_n^+ \; AR^- \xrightarrow{K_t} M_n + H^+ AR^-$$

$$HM_n^+ \; AR^- + M \xrightarrow{K_{tr}} M_n + HM^+ AR^-$$

rate of initiation:-

$$R_i = K k_i [A] [RH] [M]$$

rate of termination,

$$R_t = k_t [M^+] \quad \text{where } M^+ = HM^+ AR^-$$

Assuming steady-state approximation, we may write $R_i = R_t$ and from the relation —

$$[M^+] = \frac{K k_i [A] [RH] [M]}{k_t}$$

The overall rate of polymerisation R_p is given by —

$$R_p = k_p[M][M^+]$$

$$= K \frac{k_i k_p}{k_t} [A][RH][M]$$

If termination predominates over transformer then —

$$\bar{x} = \frac{R_p}{R_t}$$

$$= \frac{k_p[M][M^+]}{k_t[M^+]}$$

$$= \frac{k_p}{k_t}[M]$$

But when transfer predominates

$$\bar{x} = \frac{R_p}{R_t}$$

$$= \frac{k_p[M][M^+]}{k_{tr}[M][M^+]}$$

$$= \frac{k_p}{k_{tr}}$$

Lec

⊛ kinetics of anionic chain polymerisation:

1. Initiation:

$$IA \xrightarrow{k_d} I^+ A^- \quad \text{[ion pair formation]}$$

$$I^+ A^- + M \xrightarrow{k_i} AM^- I^+$$

The rate of initiation is given by —

$$R_i = -\frac{d[M_i]}{dt} = k_i [IA][M] \quad \text{——(i)}$$

2. Propogation:

$$AM^- I^+ + M \xrightarrow{k_p} AMM^- I^+$$

The rate of propogation is —

$$R_p = -\frac{d[M_p]}{dt} = k_p [M^-][M] \quad \text{——(ii)}$$

3. Termination:

$$AM_n M^- I^+ + HA \xrightarrow{k_t} AM_n MH + IA$$

The rate of termination —

$$R_t = -\frac{d[M_t]}{dt} = k_t [M^-][HA] \quad \text{——(iii)}$$

where $AM_nM^- = M^-$

By applying steady state approximation

$$k_t [M^-] [HA] = k_i [IA] [M]$$

$$\Rightarrow [M^-] = \frac{k_i [IA] [M]}{k_t [HA]}$$

Substituting these value in eqn (ii)

$$R_p = k_p [M] [M^-]$$

$$= k_p [M] \frac{k_i [IA] [M]}{k_t [HA]}$$

$$= \frac{k_i k_p}{k_t} \cdot \frac{[IA] [M]^2}{[HA]}$$

This is the rate expression for anionic chain polymerisation.

Average kinetic chain length

$$\bar{x}_n = \frac{R_p}{R_t}$$

$$= \frac{k_p [M^-] [M]}{k_t [M^-] [HA]}$$

$$= \frac{k_p}{k_t} \cdot \frac{[M]}{[HA]}$$

2016

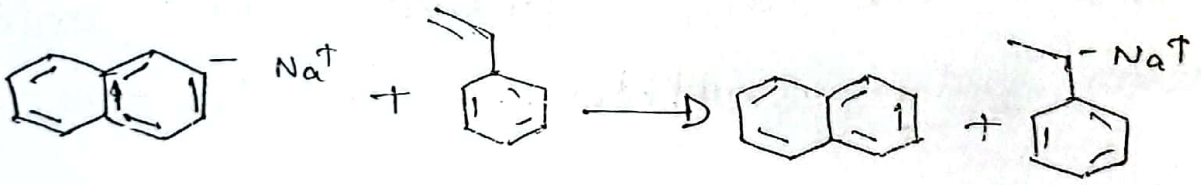
Living polymerisation:

The polymerisation in which no termination step and no chain transfer involved. Polymerisation stop when there in no more monomer. Once the monomer is replenish, polymerisation resumed.

Advantage: 1) control polydispersity
2) Molecular weight
3) Block co-polymer

Example: Anionic polymerisation sodium dihydronaphthylide with styrene

initiation:



propogation:

